METASPLOITABLE2 EXPLOITATION TECHNIQUE DOCUMENTATION

Target System:

Metasploitable2
The target system used in this internship task is the Metasploitable2 vulnerable virtual machine, which is intentionally designed for penetration testing practice.

Tool:

Metasploit Framework
The Metasploit Framework is an advanced penetration testing platform used by security professionals to develop, test, and execute exploits against vulnerable systems.

Exploit Module:

exploit/unix/ftp/vsftpd_234_backdoor
This exploit targets a malicious backdoor present in VSFTPD version 2.3.4. When triggered, the vulnerability allows attackers to gain unauthorized shell access to the target system.

Attack Technique Flow:

Reconnaissance
↓
Nmap Service Scan
↓
FTP Version Detection
↓
Metasploit Exploit Execution
↓
Payload Delivery
↓
Shell Access
↓

Post-Exploitation Activities
↓
Persistence Simulation

During the reconnaissance phase, an Nmap scan was conducted to identify open ports and services on the target Metasploitable2 system. The scan revealed that the FTP service was running VSFTPD version 2.3.4.

This version contains a known backdoor vulnerability that can be exploited using the Metasploit Framework. The exploit module exploit/unix/ftp/vsftpd_234_backdoor was selected and configured with the target IP address.

Once the exploit was executed, a payload was delivered that triggered the backdoor service and opened a remote command shell on the compromised system.

After successful access was obtained, post-exploitation techniques were performed to enumerate system information and verify the level of access gained by the attacker.

To simulate advanced attacker behavior, persistence techniques were discussed to demonstrate how a threat actor could maintain long-term access to the compromised system even after the initial exploitation phase.

Metasploit Execution Commands:

```
msfconsole

search vsftpd

use exploit/unix/ftp/vsftpd_234_backdoor

set RHOST 192.168.56.101

run
```

These commands load the exploit module, configure the target system, and execute the attack to obtain a shell on the vulnerable host.

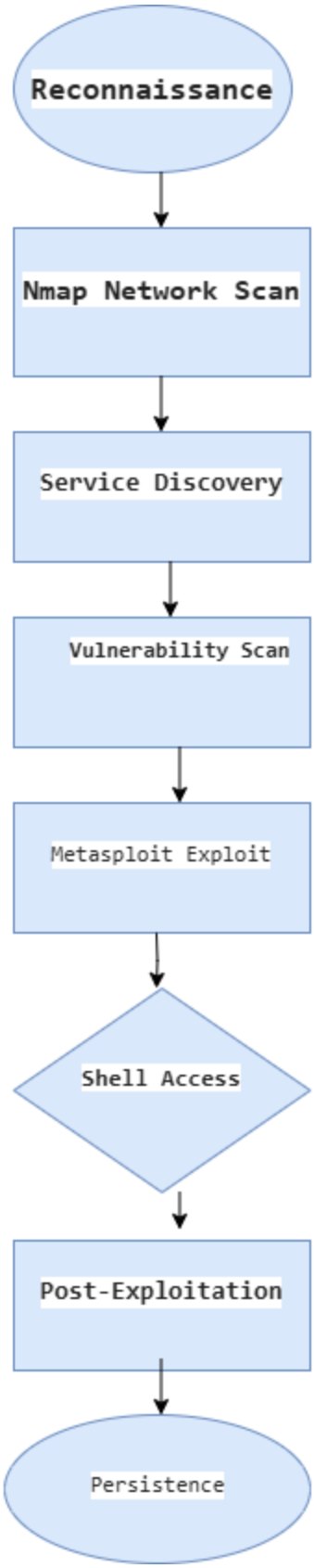
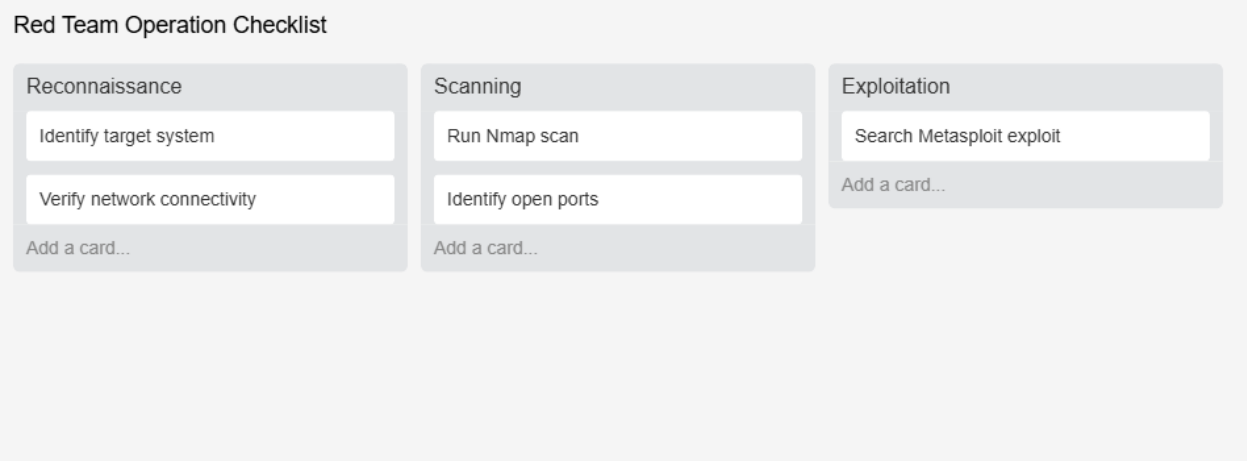
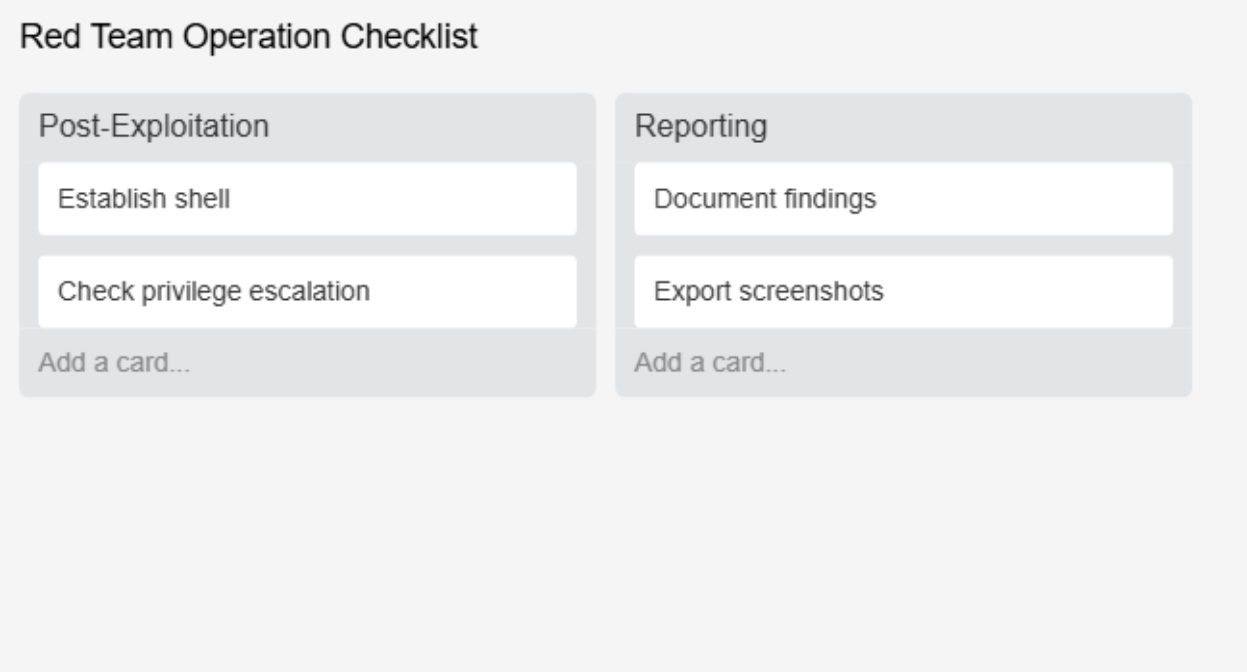
Security Impact:

Successful exploitation of this vulnerability allows attackers to gain unauthorized shell access to the system. This could result in data compromise, privilege escalation, and long-term persistence within the network.

Mitigation and Defense:

- Upgrade VSFTPD to a secure version
- Disable unnecessary FTP services
- Implement network intrusion detection systems
- Monitor suspicious login attempts

ATTACK FLOWCHART: USE DRAW.IO TO DIAGRAM AN ATTACK PATH.


Reconnaissance
Nmap Network Scan
Service Discovery
Vulnerability Scan
Metasploit Exploit
Shell Access
Post-Exploitation
Persistence

TRELLO BOARD RED TEAM CHECKLIST.